

## CAPO FRASCA, Italy

WMO: 165390

Lat: 39.7399N

Lon: 8.4595E

Elev: 95

StdP: 100.19

Time Zone: 1.00 (EUC)

Period: 94-14

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | 3.7        | 5.0 | -0.4                          | 3.7 | 5.3  | 1.2 | 4.2 | 6.2  | 17.0                  | 10.2 | 15.4 | 9.6  | 4.0                      | 50   | 0.502 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.2                    | 32.1            | 23.5 | 30.1 | 24.3 | 28.6 | 24.4 | 27.1                | 28.7 | 26.2 | 27.9 | 25.4 | 27.3 | 6.3                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.6                            | 22.5 | 28.3 | 25.7 | 21.3 | 27.4 | 24.9 | 20.2 | 26.7 | 86.0          | 28.9 | 82.0 | 28.1 | 78.5 | 27.3 | 30.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 14.5              | 12.8 | 11.1 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |      |
|                   |      |      | DB                         |     |                    |     |                                                    |     |            |      |            |      |            |      |      |
|                   |      |      | WB                         | 1.5 | 27.7               | 1.6 | 1.5                                                | 0.3 | 28.8       | -0.6 | 29.7       | -1.5 | 30.5       | -2.7 | 31.6 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.0        | 10.4  | 10.1  | 11.7  | 14.0  | 17.3  | 21.6  | 24.4  | 25.6  | 22.0  | 19.5  | 15.1  | 11.8  | (6)  |
|                                                                        | DBStd       | 5.81        | 2.27  | 2.45  | 2.14  | 1.78  | 2.14  | 2.56  | 2.23  | 2.24  | 2.14  | 2.42  | 2.70  | 2.33  | (7)  |
|                                                                        | HDD10.0     | 70          | 22    | 26    | 9     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 10    | (8)  |
|                                                                        | HDD18.3     | 1181        | 246   | 232   | 206   | 132   | 46    | 3     | 0     | 0     | 1     | 15    | 100   | 201   | (9)  |
|                                                                        | CDD10.0     | 2628        | 35    | 28    | 61    | 119   | 228   | 349   | 447   | 484   | 361   | 294   | 156   | 67    | (10) |
|                                                                        | CDD18.3     | 697         | 0     | 0     | 0     | 1     | 14    | 102   | 188   | 226   | 112   | 51    | 3     | 0     | (11) |
|                                                                        | CDH23.3     | 3984        | 0     | 0     | 0     | 3     | 37    | 423   | 1225  | 1818  | 388   | 90    | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 937         | 0     | 0     | 0     | 0     | 7     | 100   | 275   | 499   | 51    | 5     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 5.1         | 5.7   | 5.3   | 5.0   | 5.7   | 5.1   | 5.0   | 4.8   | 4.5   | 4.8   | 4.9   | 5.5   | 5.4   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 541         | 55    | 65    | 49    | 49    | 29    | 14    | 5     | 6     | 38    | 70    | 98    | 70    | (15) |
|                                                                        | PrecMax     | 790         | 169   | 167   | 161   | 109   | 142   | 60    | 29    | 38    | 127   | 288   | 261   | 170   | (16) |
|                                                                        | PrecMin     | 164         | 2     | 0     | 0     | 3     | 0     | 0     | 0     | 0     | 0     | 4     | 12    | 2     | (17) |
|                                                                        | PrecStd     | 158         | 41    | 44    | 34    | 33    | 31    | 16    | 8     | 10    | 32    | 63    | 59    | 35    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 15.7  | 16.5  | 19.3  | 22.9  | 26.4  | 32.9  | 33.6  | 35.2  | 30.3  | 27.4  | 22.1  | 17.5  | (19) |
|                                                                        |             | MCWB        | 13.8  | 13.3  | 14.0  | 18.1  | 19.1  | 21.4  | 22.6  | 23.1  | 24.0  | 21.7  | 19.9  | 15.1  | (20) |
|                                                                        | 2%          | DB          | 14.8  | 14.8  | 16.9  | 19.4  | 24.1  | 29.1  | 31.0  | 32.9  | 28.0  | 25.5  | 20.8  | 16.5  | (21) |
|                                                                        |             | MCWB        | 13.5  | 12.6  | 13.3  | 15.7  | 18.6  | 21.0  | 23.7  | 23.9  | 23.5  | 21.5  | 18.3  | 14.6  | (22) |
|                                                                        | 5%          | DB          | 14.1  | 14.0  | 15.7  | 17.9  | 22.2  | 27.0  | 29.2  | 30.7  | 26.4  | 24.1  | 19.7  | 15.6  | (23) |
|                                                                        |             | MCWB        | 13.0  | 12.2  | 13.4  | 15.3  | 18.6  | 22.0  | 24.6  | 24.8  | 23.0  | 20.9  | 17.7  | 14.1  | (24) |
|                                                                        | 10%         | DB          | 13.4  | 13.2  | 14.9  | 16.9  | 20.9  | 25.4  | 28.0  | 29.1  | 25.3  | 23.1  | 18.7  | 15.0  | (25) |
|                                                                        |             | MCWB        | 12.3  | 11.7  | 13.0  | 14.9  | 18.3  | 21.8  | 24.7  | 25.1  | 22.3  | 20.4  | 16.7  | 13.7  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 14.6  | 14.2  | 15.3  | 18.9  | 22.6  | 25.7  | 28.1  | 28.7  | 26.4  | 23.8  | 20.4  | 15.8  | (27) |
|                                                                        |             | MCDB        | 15.0  | 15.2  | 16.5  | 21.8  | 24.6  | 27.8  | 29.3  | 30.0  | 28.1  | 25.7  | 21.4  | 16.6  | (28) |
|                                                                        | 2%          | WB          | 13.8  | 13.3  | 14.5  | 16.8  | 20.4  | 24.2  | 26.8  | 27.5  | 25.1  | 22.4  | 19.1  | 15.1  | (29) |
|                                                                        |             | MCDB        | 14.4  | 14.2  | 15.9  | 18.4  | 22.1  | 26.0  | 28.2  | 29.1  | 26.4  | 24.2  | 20.2  | 15.9  | (30) |
|                                                                        | 5%          | WB          | 13.2  | 12.7  | 13.9  | 15.8  | 19.5  | 23.3  | 25.9  | 26.6  | 23.9  | 21.5  | 18.0  | 14.4  | (31) |
|                                                                        |             | MCDB        | 13.9  | 13.6  | 15.3  | 17.2  | 21.2  | 25.0  | 27.6  | 28.3  | 25.4  | 23.4  | 19.1  | 15.4  | (32) |
|                                                                        | 10%         | WB          | 12.5  | 12.1  | 13.2  | 15.1  | 18.6  | 22.5  | 25.1  | 25.8  | 22.9  | 20.6  | 17.0  | 13.7  | (33) |
|                                                                        |             | MCDB        | 13.3  | 13.0  | 14.5  | 16.5  | 20.2  | 24.3  | 26.8  | 27.7  | 24.7  | 22.6  | 18.3  | 14.9  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.1   | 4.4   | 5.3   | 5.0   | 5.4   | 5.8   | 5.8   | 6.2   | 5.7   | 5.4   | 4.3   | 3.9   | (35) |
|                                                                        |             | MCDBR       | 4.1   | 4.9   | 6.8   | 7.0   | 8.2   | 9.4   | 8.5   | 9.2   | 7.3   | 6.6   | 5.2   | 4.3   | (36) |
|                                                                        | 5% WB       | MCWBR       | 3.3   | 3.9   | 4.8   | 5.0   | 4.7   | 5.1   | 5.0   | 5.4   | 5.0   | 4.4   | 4.2   | 3.5   | (37) |
|                                                                        |             | MCWBR       | 3.9   | 4.6   | 6.2   | 6.0   | 6.5   | 6.7   | 6.5   | 7.1   | 6.7   | 5.9   | 5.0   | 4.0   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.344 | 0.366 | 0.402 | 0.438 | 0.453 | 0.432 | 0.443 | 0.433 | 0.429 | 0.405 | 0.380 | 0.351 | (40) |
|                                                                        |             | taud        | 2.431 | 2.355 | 2.254 | 2.166 | 2.162 | 2.270 | 2.244 | 2.279 | 2.281 | 2.320 | 2.335 | 2.395 | (41) |
|                                                                        | Ebn at Noon |             | 814   | 845   | 848   | 839   | 832   | 849   | 835   | 834   | 811   | 788   | 757   | 770   | (42) |
|                                                                        |             | Edh at Noon | 83    | 102   | 125   | 145   | 149   | 134   | 136   | 128   | 120   | 104   | 90    | 80    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.97  | 2.87  | 4.24  | 5.45  | 6.55  | 7.59  | 7.67  | 6.68  | 5.01  | 3.55  | 2.22  | 1.78  | (44) |
|                                                                        | RadStd      |             | 0.17  | 0.29  | 0.42  | 0.36  | 0.48  | 0.24  | 0.23  | 0.32  | 0.17  | 0.26  | 0.17  | 0.17  | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (9 neighbors) | N/A   | N/A     | N/A    | N/A     | -0.45 | -0.59 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page